Hamilton (1989): Markov switching and the filter

A reading summary for the MS-GARCH thesis

9 October 2026

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.

The paper in one paragraph

Hamilton lets the mean growth rate of an autoregression switch between two values according to an unobserved two-state Markov chain. Because the regime is never observed, he builds a nonlinear filter that turns the observed data into probabilities of being in each regime, and shows that the likelihood falls out of the filter as a by-product, so all parameters can be estimated by maximum likelihood. Applied to quarterly US real GNP growth (1951–1984), the likelihood picks out business-cycle regimes rather than slow-moving trend regimes: roughly $+1.2\%$ growth per quarter in expansions and -0.4% in recessions. The inferred recession probabilities line up closely with NBER business-cycle dates, and a typical recession lowers the long-run level of GNP by about 3%. Linear ARIMA models fit the same data almost as well on standard diagnostics, but two tests based on the filter output favour the nonlinear model.

How to read this summary. Equations are written in the thesis notation, not Hamilton's. Hamilton's own symbols are given in the translation table (Section 10) so you can follow the PDF. Numbers come from the paper; figures I computed from his estimates are marked *(computed)*. Symbols not yet in `notation.md` are listed in Section 11.

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1 Where the paper fits in the thesis

- **§2.4 (regime switching as the alternative account).** This is the paper that introduced the Markov-switching idea to econometrics in its now-standard form. It is the reason the thesis can say that an unobserved regime can be estimated, dated and forecast.
- **§3.1 (deriving path dependence).** The filter only works because the density of the data depends on a *finite* number of lagged regimes. Path dependence in MS-GARCH is precisely the failure of that condition.
- **Chapter 4 (filtering; testing the number of regimes).** The filter, the likelihood, the expected-duration formula $1/(1 - p_{kk})$ and the smoother are all used directly. A footnote also states the Davies problem for testing one regime against two.
- **Chapter 5 (regime dating).** The comparison with NBER dates is the template for asking whether each MS-GARCH model's high-volatility regime matches known crisis episodes.

2 The question

In the 1980s, the debate about trends in GNP (Nelson–Plosser, Campbell–Mankiw, Watson, Clark) assumed that output growth follows a *linear* stationary process, so that optimal forecasts are linear in past values. Hamilton asks what happens if growth instead follows a *nonlinear* stationary process: one that is subject to occasional, discrete shifts between episodes with markedly different behaviour.

Two features distinguish his approach from earlier work:

- **Unobserved regimes.** Unlike Tong, who treated regime shifts as observable, Hamilton's econometrician never sees the regime and must infer it from the data.
- **Proper marginal likelihood.** Unlike Sclove, who maximised the joint likelihood over both parameters and an imputed regime sequence, Hamilton integrates the regimes out, maximises the marginal likelihood of the observed data over the parameters only, and then uses the estimates to compute regime probabilities.

He describes the filter as the discrete-state counterpart of the Kalman filter: both infer an unobserved state from an observed series, but the Kalman filter is linear with a continuous state, while this filter is nonlinear with a discrete state.

3 The model

3.1 Specification

Let y_t be 100 times the quarterly change in log real GNP. The regime $S_t \in \{1, 2\}$ follows a first-order Markov chain with transition probabilities $p_{ij} = P(S_t = j \mid S_{t-1} = i)$. In this summary regime 1 is the *expansion* state and regime 2 the *recession* state. Hamilton models deviations of growth from its regime-specific mean as a Gaussian AR(4):

$$y_t = \mu_{S_t} + z_t, \qquad z_t = \sum_{m=1}^{4} \phi_m z_{t-m} + \sigma \eta_t, \qquad \eta_t \sim \text{i.i.d. } N(0,1), \tag{1}$$

with η_t independent of the regime process. Substituting $z_{t-m} = y_{t-m} - \mu_{S_{t-m}}$ gives

$$y_t \mid S_t, \ldots, S_{t-4}, \mathcal{F}_{t-1} \;\sim\; N\Big(\mu_{S_t} + \sum_{m=1}^{4} \phi_m \big(y_{t-m} - \mu_{S_{t-m}}\big),\; \sigma^2\Big). \tag{2}$$

The key structural fact is visible in (2): the density of y_t depends on the current regime *and four lagged regimes*, but on nothing further back. The relevant "state" is therefore the vector $(S_t, \ldots, S_{t-4})$, which has $2^5 = 32$ possible values.

In the paper this is written as a Markov *trend* in the level of log GNP, $n_t = n_{t-1} + \mu_{S_t}$, plus a component whose first difference is the AR(4) above. Differencing gives (1). Note also that the model is not identified up to relabelling the regimes; Hamilton normalises by requiring the "fast growth" regime to have the higher mean.

3.2 Properties of the regime process

- **AR(1) representation.** The indicator of the expansion regime follows an AR(1) whose autoregressive coefficient is $p_{11} + p_{22} - 1$. This is why the thesis calls $p_{11} + p_{22} - 1$ *Markov persistence*: it governs how fast the probability of each regime decays towards its long-run value.
- **Stationary probability.** The forecast probability of each regime converges geometrically to $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$, whatever is known today.
- **Uncorrelated but not independent.** The innovation in this AR(1) is uncorrelated with past regimes but *not* independent of them: its variance depends on the previous regime.

The last point has a practical consequence. When noise is added (as in (1)), the observed growth series still has a valid ARMA representation, but forecasts that are linear in past y_t are *not* optimal. The best forecast uses the nonlinear filtered probability of the current regime.

3.3 Forecasting and permanent effects

Hamilton derives closed-form forecasts for the level of the series, both for a Markov trend in logs and in levels, together with discounted present values. Two results matter:

- The **long-run growth rate** of the forecast does not depend on what is known about today's regime; it converges to the stationary-probability-weighted mean growth.
- The **long-run level** does. Knowing that the economy is in expansion rather than recession today raises the forecast of log output at all long horizons by

$$(\mu_1 - \mu_2)\,\frac{p_{11} + p_{22} - 1}{2 - p_{11} - p_{22}}. \tag{3}$$

  So regime information has a *permanent* effect on the level, just as a shock to an ARIMA(p,1,q) process does.

4 Estimation: the filter, the likelihood and the smoother

4.1 The basic filter

Write $\mathcal{F}_t$ for the information in the observed series up to t. The filter carries forward the joint probability of the four most recent regimes and updates it each period in five steps.

1. **Predict.** Extend the input $P(S_{t-1}, \ldots, S_{t-4} \mid \mathcal{F}_{t-1})$ (16 values) to 32 values by multiplying by the transition probability:

$$P(S_t, S_{t-1}, \ldots, S_{t-4} \mid \mathcal{F}_{t-1}) = p_{S_{t-1}S_t}\, P(S_{t-1}, \ldots, S_{t-4} \mid \mathcal{F}_{t-1}).$$

2. **Joint density.** Multiply each of the 32 probabilities by the Gaussian density (2).
3. **Likelihood.** Sum over all 32 configurations to get $f(y_t \mid \mathcal{F}_{t-1})$.
4. **Update.** Divide step 2 by step 3 (Bayes' rule) to get $P(S_t, \ldots, S_{t-4} \mid \mathcal{F}_t)$.
5. **Collapse the oldest lag.** Sum out S_{t-4} to obtain $P(S_t, \ldots, S_{t-3} \mid \mathcal{F}_t)$, the input for the next period.

Marginalising the output gives the filtered probability $\xi_{k,t|t} = P(S_t = k \mid \mathcal{F}_t)$. Every quantity is an exact probability, so the output always lies in $[0, 1]$; Hamilton contrasts this with the continuous-time filter of Liptser and Shiryayev, which can produce values outside the unit interval when used as an approximation.

Starting values. Hamilton starts the filter from the unconditional (stationary) probabilities of the regime vector rather than computing the exact conditional distribution of the initial regimes. If a regime is absorbing (e.g. a permanent break), the initial probability should instead be estimated as a parameter.

4.2 The likelihood

The log-likelihood is the sum of the step-3 terms,

$$\ell_T(\theta) = \sum_{t=1}^{T} \ln f(y_t \mid \mathcal{F}_{t-1}), \qquad \theta = (\mu_1, \mu_2, p_{11}, p_{22}, \sigma, \phi_1, \ldots, \phi_4),$$

which Hamilton maximises numerically (Davidon–Fletcher–Powell), with standard errors from numerical second derivatives. He reports that the maximum was robust to a wide range of starting values.

4.3 Generalisations noted in the paper

The same logic works with more than two regimes (the state vector then has K^5 elements), with switching autoregressive coefficients, with a switching innovation variance (done in Hamilton's 1988 paper, not in the folder), and with a higher-order Markov chain. Hamilton notes the limits are mainly numerical: identifying the regime dynamics separately from the Gaussian dynamics relies on nonlinearity in the data, so richness on both sides cannot be pushed far.

4.4 Smoothing

- **r-lag smoother.** Because the filter already carries $(S_t, \ldots, S_{t-3})$, it delivers $P(S_{t-4} \mid \mathcal{F}_t)$ for free: an inference about the regime four quarters ago using data up to today.

- **Full-sample smoother.** Adapting Cosslett and Lee (1985), Hamilton obtains $\xi_{k,t|T}$ by re-running the filter from each date for each configuration of regimes. This is computationally heavy; Kim (1994) later provided the efficient backward recursion that is used in practice.
- **Result.** The four-lag and full-sample smoothers differ very little in the GNP data (average absolute difference 0.016), so the cheaper four-lag smoother is often enough.

The point for Chapter 3: why the filter needs finite lags

The filter is feasible because the state that matters, $(S_t, \ldots, S_{t-4})$, has a *fixed* number of elements (2^5). Every regime model in the thesis is judged by whether it preserves this property:

- **SWARCH** (Hamilton & Susmel, 1994) keeps it: with ARCH(q) the state is $(S_t, \ldots, S_{t-q})$, i.e. K^{q+1} configurations.
- **Path-dependent MS-GARCH** loses it: the lagged variance carries the whole history, so the "state" would be $S_{1:t}$, with K^t configurations.
- **IS-GARCH** keeps it in the simplest form: each regime variance is a function of observed data only, so the state is just S_t (K configurations).

5 Empirical results: US real GNP

5.1 Data and estimates

Quarterly US real GNP (1982 dollars), with growth rates used from 1951:II and the filter started in 1952:II to allow four lags.

Thesis symbol	Meaning	Estimate	Std. error
μ_2	mean growth, recession	-0.358	0.265
$\mu_1 - \mu_2$	extra growth in expansion	1.522	0.264
μ_1	mean growth, expansion	1.164 *(computed)*	—
p_{11}	stay in expansion	0.9049	0.0374
p_{22}	stay in recession	0.7550	0.0966
σ	innovation std. dev.	0.769	0.067
$\phi_1, \ldots, \phi_4$	AR coefficients	0.014, -0.058, -0.247, -0.213	0.11–0.14

Derived quantities *(computed from the table)*:

- Markov persistence $p_{11} + p_{22} - 1 \approx 0.66$.
- Stationary probability of expansion $\pi_1 \approx 0.72$.
- Expected durations $1/(1 - p_{kk})$: about 10.5 quarters for expansions and 4.1 for recessions. Hamilton reports both and compares them with NBER averages of 14.3 and 4.7 quarters.

5.2 Interpretation

- **Business cycles, not trends.** One might have expected the regimes to separate decades of faster and slower trend growth. Instead the likelihood is maximised by short-lived regimes with negative growth in one and positive growth in the other, i.e. business-cycle phases.
- **Regimes replace short-lag autocorrelation.** ϕ_1 and ϕ_2 are close to zero: the positive serial correlation of GNP growth at short lags is captured by the persistence of the regimes rather than by autoregressive coefficients. Hamilton suggests the negative third and fourth lags may be an artifact of seasonal adjustment applied to a nonlinear process.

5.3 Dating business cycles

Using the full-sample smoother and the rule "recession if $\xi_{2,t|T} > 0.5$", Hamilton dates postwar peaks and troughs and compares them with the NBER.

- For most recessions the two datings are within three months of each other.
- The largest differences are 1957–58 and 1979–80, where the smoother places the recession start immediately after oil price increases (the Suez crisis and the Iranian revolution).
- Very few smoothed probabilities lie between 0.3 and 0.7, so the 0.5 threshold hardly matters. Hamilton reads this as evidence that the filter is detecting a genuine two-phase pattern rather than fitting an arbitrary nonlinearity.

Why it matters for the thesis

The dating exercise is the model for regime dating in Chapter 5: report the smoothed probabilities, compare high-volatility episodes with known events, and check how *decisive* the probabilities are. A model whose smoothed probabilities cluster near 0 and 1 is giving a sharper regime classification than one with many intermediate values.

6 Linear versus nonlinear models

6.1 Why linear models fit so well

Hamilton asks why parsimonious ARIMA models seemed adequate if his model is the truth.

- **Autocorrelations.** The theoretical autocorrelation function implied by the estimated Markov model is hard to distinguish from that of a low-order ARIMA in a sample of postwar size.
- **Monte Carlo.** In 1,000 samples of 130 quarters simulated from the Markov model, an AR(4) fitted by OLS gives average coefficients very close to those estimated on actual GNP, and its residuals show negligible autocorrelation.

So the success of linear models on standard diagnostics is exactly what one would expect if the Markov model generated the data (the "encompassing" argument).

6.2 Two tests that discriminate

Both tests use the lagged filter output, $\xi_{1,t-1|t-1}$, as a regressor.

- **Conditional mean.** If the Markov model is right, adding $\xi_{1,t-1|t-1}$ to an AR(4) for GNP growth should give a positive coefficient (expected forecast difference between expansion and recession of about 1%). On actual data the coefficient is 1.670 with a t-statistic of 2.83.
- **Conditional heteroskedasticity.** The Markov model implies that AR(4) forecast errors are smaller after an expansion than after a recession (predicted difference in squared errors about -0.229). Regressing squared AR(4) residuals on a constant and the lagged filter output gives a coefficient of -0.813. The Breusch–Pagan statistic is 5.17 and Engle's TR^2 is 4.75, both above the 5% χ^2_1 critical value of 3.84. Squared residuals are about twice as large after a confidently identified recession.

Hamilton cautions that the filter output is a *generated regressor*, so the right distribution for these statistics is unclear.

6.3 Why not a likelihood-ratio test?

A footnote explains why the natural test (an AR(4) as the special case $\mu_1 = \mu_2$) cannot be done in the usual way:

- under the null the transition probabilities are not identified, so their estimates converge to nothing;
- the information matrix is singular under the null;
- the score with respect to $\mu_1 - \mu_2$ is zero at the restricted estimate.

He points to Davies (1977) and others for possible remedies.

Why it matters for the thesis

This footnote is the earliest statement in the folder of the problem Chapter 4 calls the *Davies problem*: one regime against two cannot be tested with a standard LR, Wald or LM test, because some parameters exist only under the alternative. It supports the thesis decision to rely mainly on information criteria for the number of regimes.
Note also the conditional heteroskedasticity result: regime switching in the *mean* already produces ARCH-like behaviour in residuals from a linear model. This is a small, early instance of the thesis's broader theme that unmodelled regimes can masquerade as other dynamics.

7 Long-run consequences of recessions

- **Permanent level effect.** Plugging the estimates into (3), being in recession rather than expansion lowers the long-run forecast of (100 times log) GNP by about 3%. Computing the same quantity for the level rather than the log of GNP gives virtually the same answer.
- **Permanent income.** With a quarterly discount factor of 0.99, certain knowledge that a recession has begun lowers the present value of future output by about 3% (ratio 1.029).
- **Comparison with linear measures.** Earlier studies measured persistence by the long-run effect of a 1% output innovation in a linear model, with estimates between about 0.55% and 1.68%. The linear (Wold) representation of Hamilton's model gives 1.62, and this figure is almost entirely driven by the regime component. The non-regime Gaussian shocks on their own have a long-run effect of only about 0.66 *(computed as* $1/(1 - \sum_m \phi_m)$*).*

Why it matters for the thesis

There is a parallel worth drawing in §2.4, phrased carefully because it is an analogy rather than a claim made by Hamilton: in his model, a *linear* persistence measure is dominated by the contribution of the *regimes*, not by ordinary shocks. The thesis asks the same question of volatility: whether the single-regime GARCH persistence $\alpha + \beta$ is similarly dominated by unmodelled regime shifts.

8 Limits for use in the thesis

- **Mean switching, constant variance.** Regimes shift the mean growth rate; the innovation variance is constant. Volatility regimes are a later development (Hamilton's 1988 paper switches σ; Cai and Hamilton & Susmel switch ARCH variances).
- **Data.** Quarterly macroeconomic data with about 130 observations, not daily returns. The regimes are fairly short-lived ($p_{11} + p_{22} - 1 \approx 0.66$), much less persistent than the volatility regimes in the MS-GARCH papers, where staying probabilities typically exceed 0.97.

- **Inference.** Standard errors come from a numerical Hessian, and no formal test of one regime against two is available.
- **Smoother.** The full-sample smoother presented here is superseded in practice by Kim (1994).

9 Connections to other papers in the folder

Paper	Connection to Hamilton (1989)
Kim (1994)	Extends the framework to general state-space models; efficient smoother; collapsing of regime histories.
Cai (1994)	Applies the same filter with regime switching in the ARCH intercept; explains why GARCH breaks it.
Hamilton & Susmel (1994)	SWARCH: the filter with K^{q+1} configurations; same dating exercise for stock volatility.
Gray (1996), Klaassen (2002)	Restore a finite state by averaging the lagged variance over regimes.
Haas et al. (2004)	IS-GARCH: the state reduces to S_t alone, so Hamilton's filter applies directly.
Davies (1977), Cho & White (2007)	Address the testing problem stated in Hamilton's footnote.
Ardia et al. (2019)	MSGARCH returns the filtered, predicted and smoothed probabilities used for dating.

10 Translation table

Hamilton (1989)	Meaning	Thesis notation (this summary)
$s_t \in \{0, 1\}$	regime; 1 = fast growth, 0 = slow growth	$S_t \in \{1, 2\}$; 1 = expansion, 2 = recession
p	$P(s_t = 1 \mid s_{t-1} = 1)$	p_{11}
q	$P(s_t = 0 \mid s_{t-1} = 0)$	p_{22} (**not** our ARCH order q)
$\lambda = -1 + p + q$	AR(1) coefficient of the regime	$p_{11} + p_{22} - 1$ (no symbol)
π	limiting probability of $s_t = 1$	π_1
a_0	mean growth in state 0	μ_2
$a_0 + a_1$	mean growth in state 1	μ_1
$\phi_1, \ldots, \phi_r$; r	AR coefficients; AR order	ϕ_m; order 4
ε_t, σ	Gaussian AR innovation and its std. dev.	$\sigma\eta_t$
y_t	100 × change in log GNP	y_t
$P[S_t = 1 \mid y_t, y_{t-1}, \ldots]$	filtered probability	$\xi_{1,t\|t}$
$P[S_t = 0 \mid y_T, \ldots]$	smoothed recession probability	$\xi_{2,t\|T}$

Two pitfalls when reading the PDF: Hamilton's q is a transition probability, and his state numbering (0 = recession) is the reverse of the "regime 1 is the first state" convention. His ε_t is a Gaussian innovation in the output equation, not the thesis's return innovation.

11 Symbols used here that are not in notation.md

These are needed only to describe Hamilton's model. Add them to `notation.md` only if §2.4 or Chapter 4 writes the model out.

- **NEW SYMBOL** y_t: a generic observed series (here GNP growth), to keep it distinct from returns r_t.
- **NEW SYMBOL** μ_k: mean of y_t in regime k (already proposed as optional for Cai and Gray).

- **NEW SYMBOL** ϕ_m: lag-m autoregressive coefficient of the mean equation, $m = 1, \ldots, 4$. Uses the approved lag index m.
- **NEW SYMBOL** z_t: deviation of y_t from its regime mean.
- **NEW SYMBOL** σ (no time subscript): constant innovation standard deviation. If it clashes with σ_t in the thesis text, write σ_z instead.

Reference

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica, 57*(2), 357–384. https://doi.org/10.2307/1912559